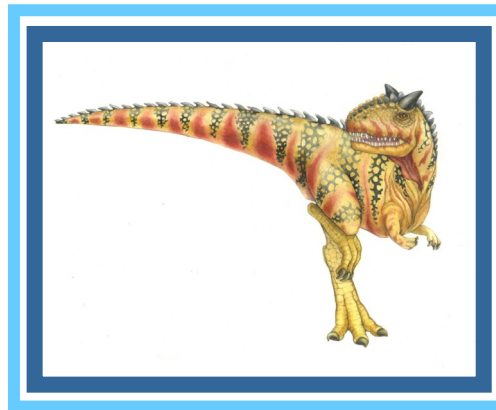
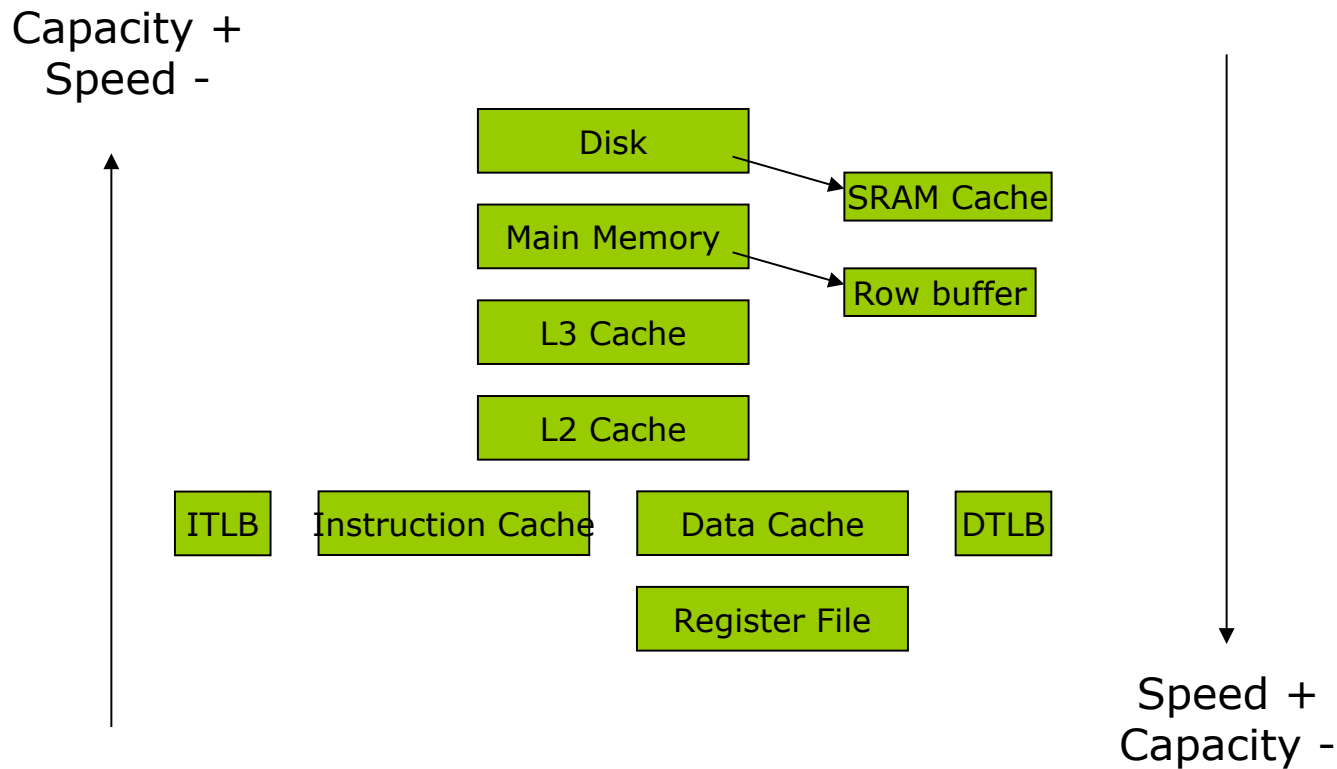


Chapter 8: Memory Management





Storage Hierarchy and Locality





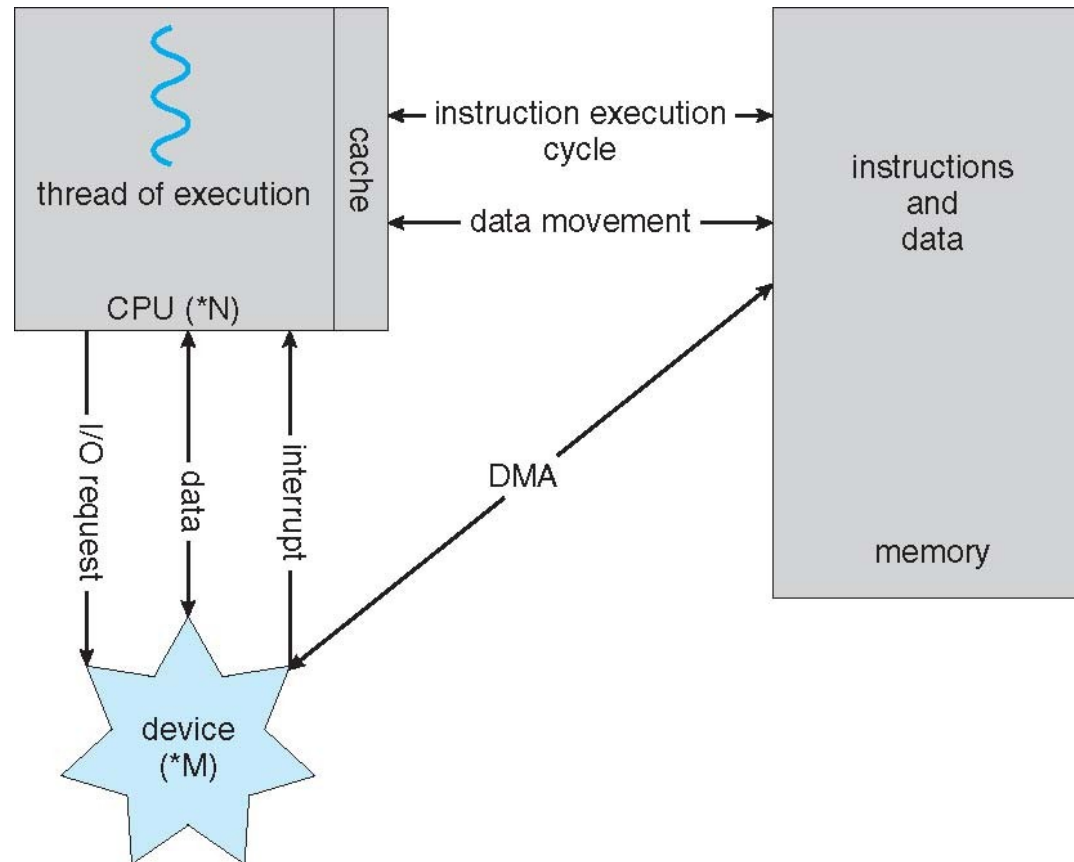
Chapter 8: Memory Management

- Background
- Contiguous Memory Allocation
- Segmentation
- Paging
- Swapping
- Virtual Memory
- Page Faults
- Page and Frame Replacement algorithms





How a Modern Computer Works

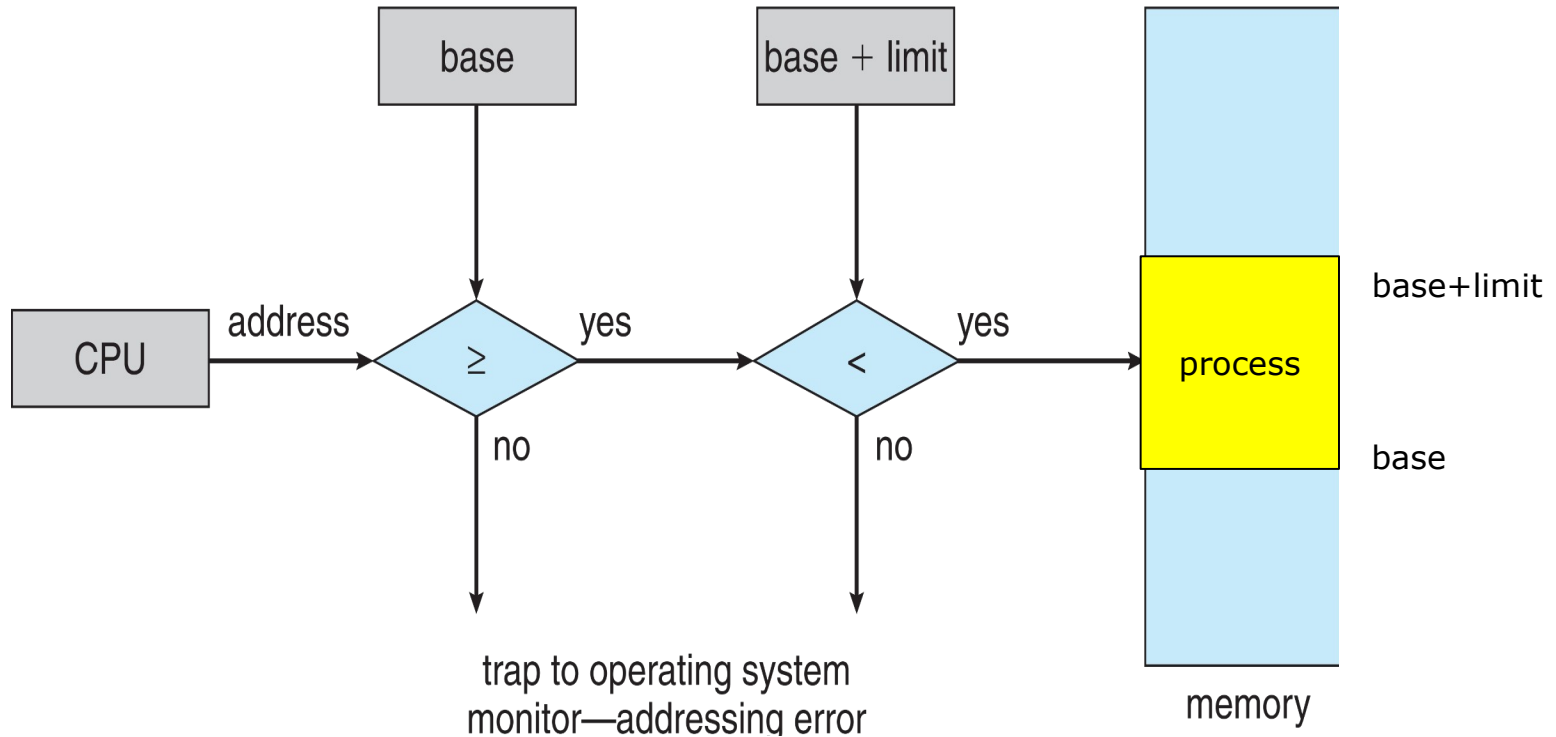


A von Neumann architecture





Hardware Address Protection





Contiguous Allocation

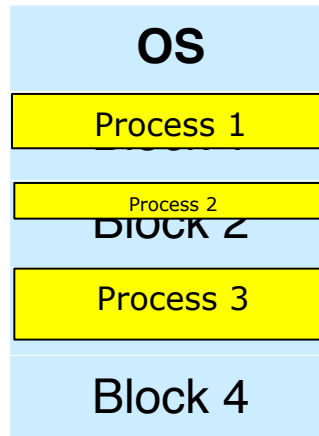
- Main memory must support both OS and user processes
- Limited resource, must allocate efficiently
 - *How?*
- Contiguous allocation is one early method
- Main memory usually into two **partitions**:
 - Resident operating system, usually held in low memory with interrupt vector
 - User processes then held in high memory
 - Each process contained in single contiguous section of memory





Multiple-partition allocation

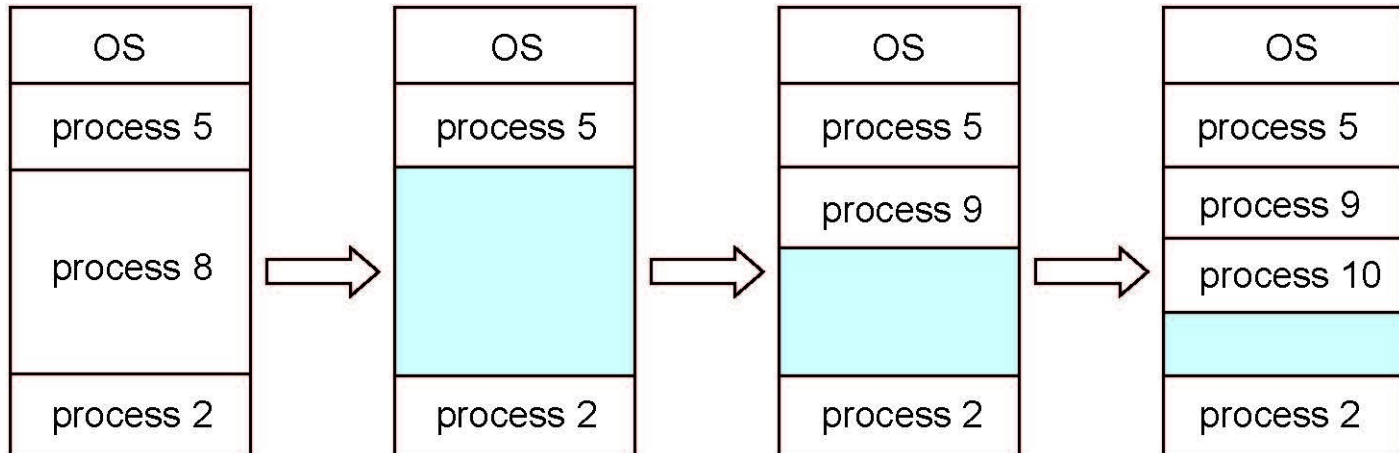
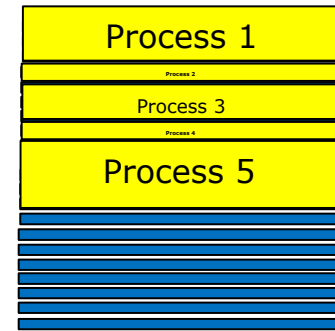
- Multiple-partition allocation
 - Fixed size partitions in the main memory
- *Any limitation?*
 - Degree of multiprogramming limited by number of partitions
 - Not efficient, not flexible





Variable-partition allocation

- **Variable-partition** allocation
 - Sized to a given process' needs for efficiency
 - ▶ *How?*
 - **Actually it is very small sized partitions**
 - **Hole** – block of available memory; holes of various size are scattered throughout memory
 - When a process arrives, it is allocated memory from a hole large enough to accommodate it
 - Process exiting frees its partition, adjacent free partitions combined
 - Operating system maintains information about:
 - a) allocated partitions b) free partitions (hole)





Dynamic Storage-Allocation Problem

How to satisfy a request of size n from a list of free holes?

- **First-fit**: Allocate the **first** hole that is big enough
- **Best-fit**: Allocate the **smallest** hole that is big enough; must search entire list, unless ordered by size
 - Produces the smallest leftover hole
- **Worst-fit**: Allocate the **largest** hole; must also search entire list
 - Produces the largest leftover hole

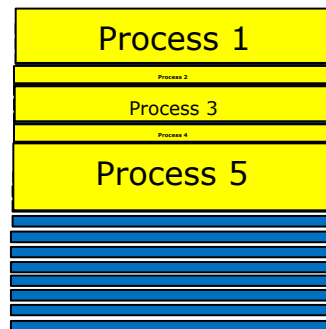
First-fit and best-fit better than worst-fit in terms of speed and storage utilization





Fragmentation

- ❑ **External Fragmentation** – total memory space exists to satisfy a request, but it is not contiguous
- ❑ **Internal Fragmentation** – allocated memory may be slightly larger than requested memory; this size difference is memory internal to a partition, but not being used
- ❑ First fit analysis reveals that given N blocks allocated, $0.5 N$ blocks lost to fragmentation
 - ❑ $1/3$ may be unusable -> **50-percent rule**



External fragmentation



Internal fragmentation





Defragmentation (Cont.)

- Reduce external fragmentation by **compaction or defragmentation**
 - Shuffle memory contents to place all free memory together in one large block
 - Compaction is possible *only* if relocation is dynamic, and is done at execution time
 - Time consuming and all the processes are affected
- Now consider that backing store has same fragmentation problems
- *Any other methods for defragmentation?*
 - Segmentation
 - Paging





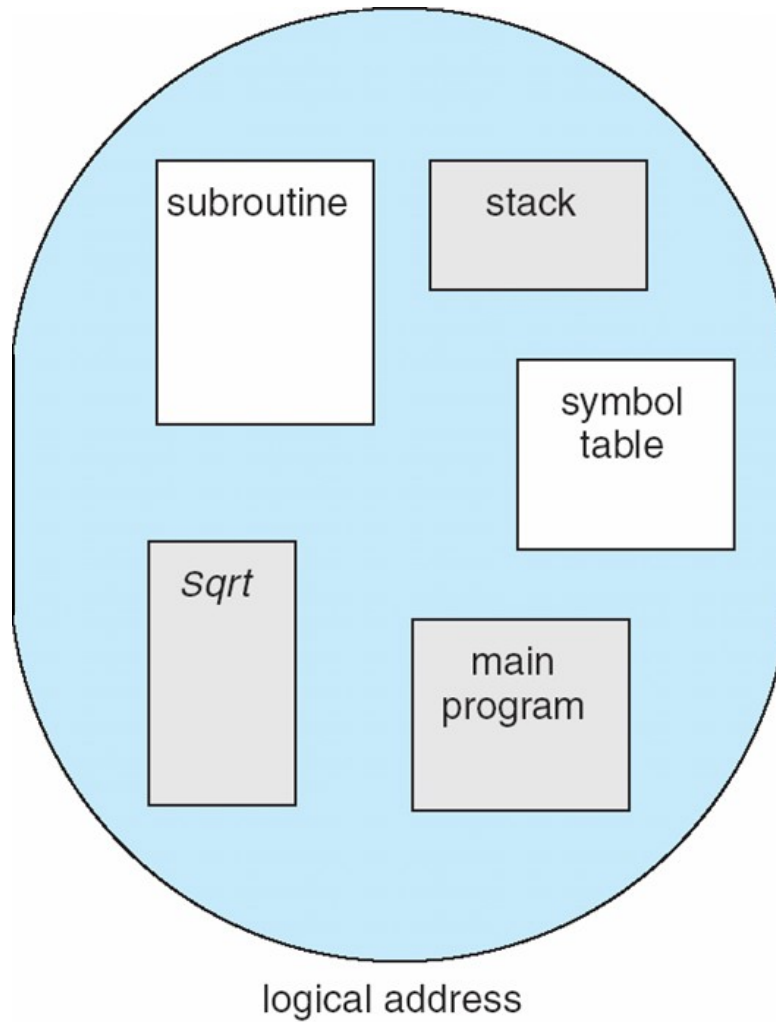
Segmentation

- Memory-management scheme that supports user view of memory
- A program is a collection of segments
 - A segment is a logical unit such as:
 - main program
 - procedure
 - function
 - method
 - object
 - local variables, global variables
 - common block
 - stack
 - symbol table
 - arrays



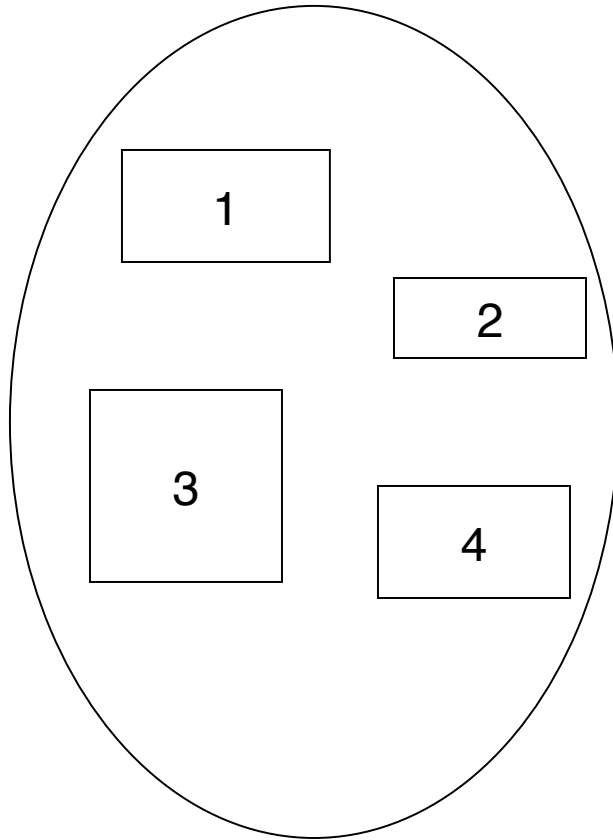


User's View of a Program

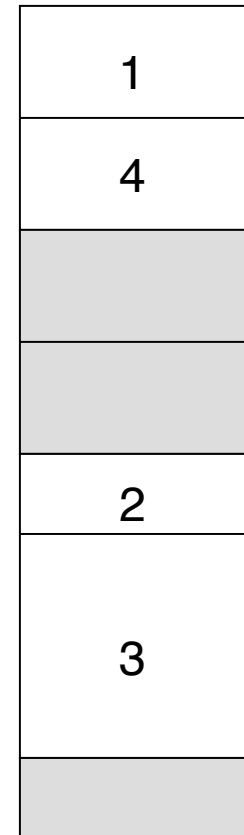




Logical View of Segmentation



user space

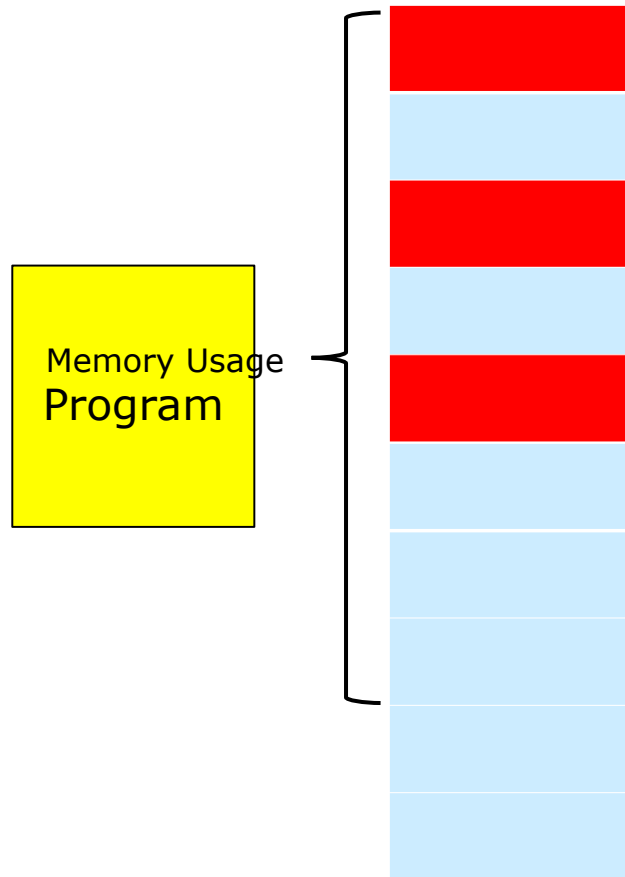


physical memory space

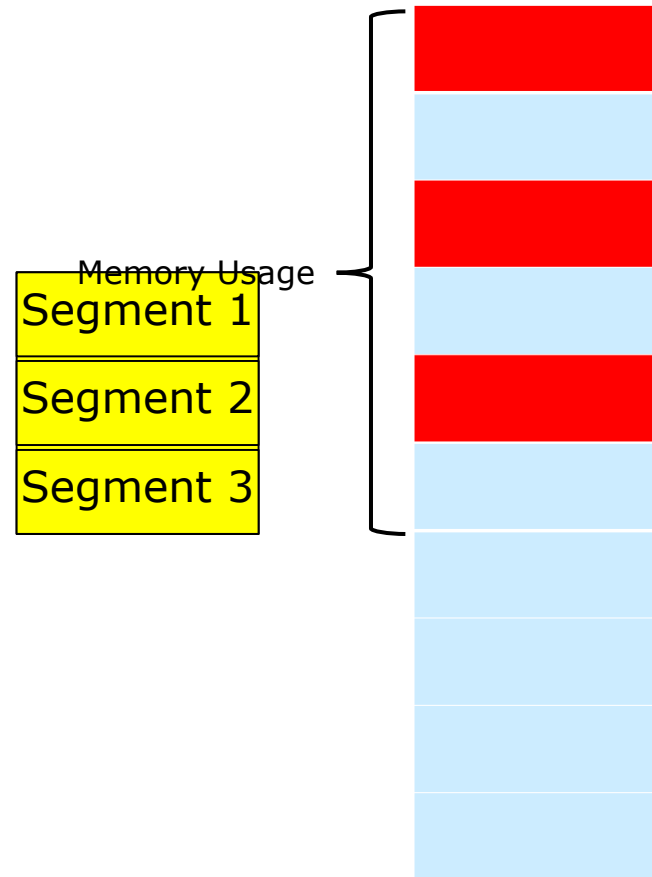




Visualization of Segmentation



No Segmentation



Segmentation

